

Ratio & Proportion

Quick reference sheet · GCSE Maths, Ratio, Proportion & Rates of Change · keep handy for revision

1. Simplifying ratios

A ratio compares two or more quantities, written with a colon (e.g. 3:5). Simplify by dividing every part by their highest common factor — just like simplifying a fraction.

15:20 → divide both by 5 → **3:4**

If the quantities have different units, convert to the same unit first: 2 kg : 500 g → 2000 g : 500 g → **4:1**

A ratio can be written in the form **1:n** by dividing both sides by one part: 8:20 → divide by 8 → **1:2.5**

2. Sharing in a given ratio

1. Add the ratio parts to find the **total number of parts**.
2. Divide the amount by the total parts to find **what one part is worth**.
3. Multiply that back up for each share.

Share £60 in the ratio 2:3:5 → total parts = 10 → 1 part = £6 → shares = **£12 : £18 : £30**

Working backwards: if you know one share, divide by its number of parts to find 1 part, then use that to find the total.

Ratio 3:4, larger share = £48 → 4 parts = £48 → 1 part = £12 → total (7 parts) = **£84**

3. Direct proportion

Two quantities are in **direct proportion** when they increase (or decrease) at the same rate: double one, the other doubles too. Written as **$y = kx$** . Most problems use the **unitary method**: find the value of 1 unit, then scale up.

5 pens cost £3.50 → 1 pen = £0.70 → 8 pens = $8 \times £0.70 =$ **£5.60**

Best buy: find the price per single item (or per 100g/litre) for each option, then compare. Shop A: 6 apples for £2.40 (£0.40 each). Shop B: 10 apples for £3.80 (£0.38 each) → **Shop B** is better value.

4. Inverse proportion

Two quantities are in **inverse proportion** when one increases as the other decreases at a matching rate: double one, the other halves. Written as **$y = k \div x$** (equivalently $xy = k$, a constant).

4 workers take 15 hours to build a wall → total worker-hours = $4 \times 15 = 60$ → 6 workers: $60 \div 6 =$ **10 hours**

y is inversely proportional to x. When $x = 5$, $y = 12$ → $k = xy = 60$ → when $x = 3$, $y = 60 \div 3 =$ **20**

Spotting the difference: quantities changing together (both up or both down) → direct. One up while the other goes down → inverse.